

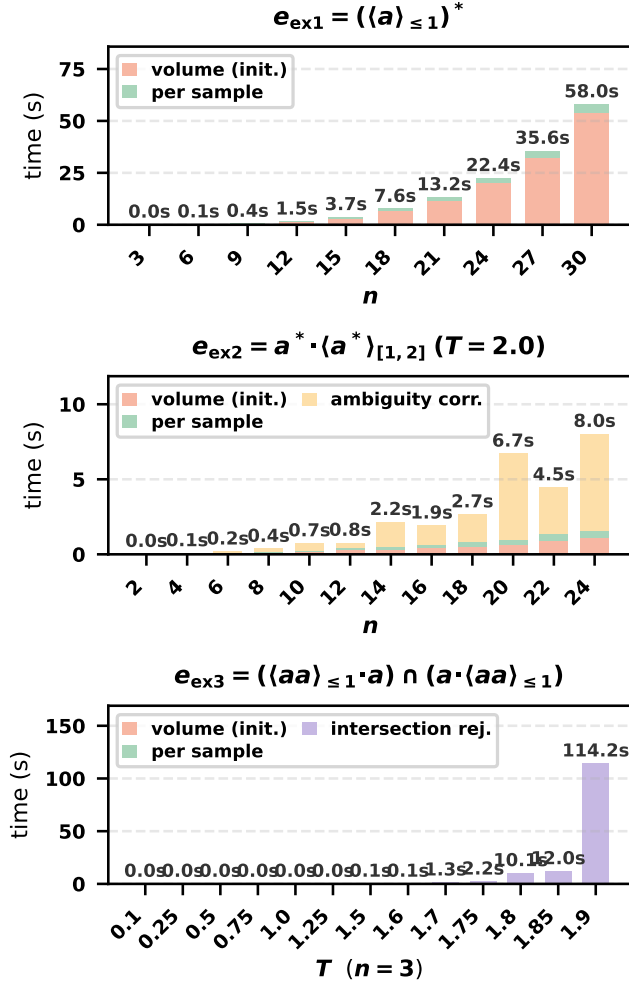


Fig. 1. Stress-test timing for three TREs ($K = 10$ samples per data point, mean sampling time in seconds). Bars decompose into one-time volume-computation cost and per-sample cost. Top: $e_{ex1} = ((a) \leq 1)^*$. Middle: $e_{ex2} = a^* \cdot (a^*)_{[1,2]} (T = 2.0)$. Bottom: $e_{ex3} = ((aa) \leq 1 \cdot a) \cap (a \cdot (aa) \leq 1) (n = 3)$.

I. Applications and Experiments

A. Scalability

Fig. 1 characterises scaling across three TREs, each stressing a different aspect of the algorithms ($K = 10$ samples per data point). Timing decomposes into a one-time volume computation (Thm. ??), paid once per specification regardless of how many words are subsequently drawn, and a per-sample cost. In typical use, the former is amortised across all samples.

For the unambiguous expression e_{ex1} , Alg. ?? proceeds without rejection and scales to $n = 22$ with cost dominated by volume computation. For the ambiguous e_{ex2} , volume computation remains inexpensive, while per-sample cost grows with n , consistent with Alg. ?? incurring an increasing density correction cost as the average multiplicity of generated words grows with n (Thm. ??). For the intersection expression e_{ex3} , Alg. ?? draws candidates from the first subexpression and accepts those also belonging

to the second. As T approaches the feasibility boundary at $T = 2$ from below, the ratio of the intersection volume to the enclosing language volume vanishes, so the expected cost per accepted sample grows accordingly. This is the expected behaviour of rejection sampling as the acceptance probability decreases.

The density correction cost in Alg. ?? can be significant for highly ambiguous expressions. The expected number of trials is characterised by the multiplicity structure of the expression (Thm. ??). The rejection-based handling of intersection in Alg. ?? is a known limitation of the approach: one can construct TREs with arbitrarily thin intersections, and the cost scales with the inverse of the acceptance probability.